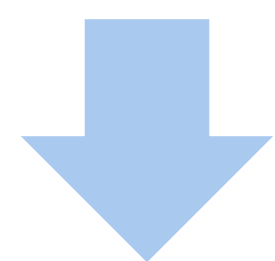
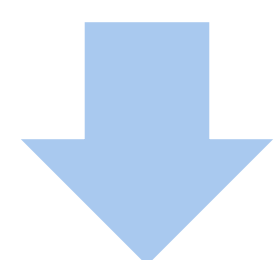


## BUILD

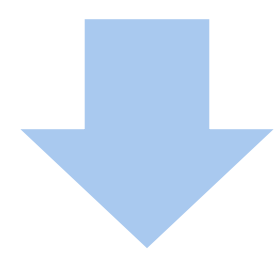
Discharge notes  
328,839 MIMIC-IV



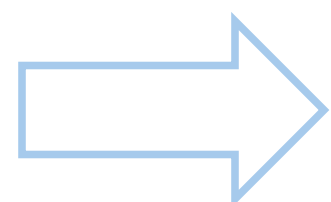
Qwen extraction  
cause → effect triples



UMLS normalization  
merge synonyms to CUIs



Candidate causal KG  
4.3K concepts / 19K candidates



## AUDIT

### Which validation signals deserve trust?

SemMedDB is an independent literature reference, not causal ground truth.

**WORKS**

#### Co-occurrence lift

EHR support signal  
filters weakly supported co-mentions

**FAILS**

#### Coding-order time

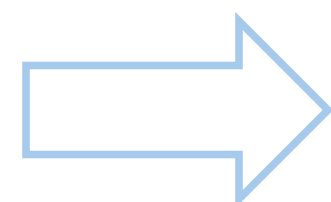
Weak direction signal  
54.2% vs 69.4%; coding order is not onset

**PARTIAL**

#### Literature evidence

Independent reference  
74% direction agreement; partial coverage

Takeaway: lift supports candidates;  
coding-order time should not orient candidates.



## APPLY

EHR-supported  
causal-candidate KG



Support-aware causal  
RAG + demotion



Held-out patient  
causal question

Unsupported hallucinations  
17.3% → 6.1%